Youssef Eslam Hussein Assignment 4 System Programming

① Op code n i x b p e disp

000000 1 1 1 0 1 0 0100 1101 0000

03A4D0

TA = disp + x + pc = 4D0 + 0030 + 2000 = 2500 Value = TA = 2500 = 000ABC

② Op code n i x b p e Address

000000 1 1 0 0 0 1 0000 1000 1001 1010 1011

031089AB

TA = Address = 089AB Value = (TA) = 089AB → 00C303

③ Op code n i x b p e disp

000000 0 1 0 0 0 0 0001 0101 0101

010155

TA = disp = 155 Value = TA = 000155

④ Op code n i x b p e disp

000000 1 0 0 0 1 0 0101 0000 0000

022500

TA = disp + pc = 500 + 2000 = 2500 Value = ((TA)) = ((2500)) = (ABC) = 006060

⑤ Op code n i x b p e Address

000000 1 0 0 0 0 1 0000 0000 1010 1011 1100

02100ABC

TA = address = ABC Value = ((TA)) = ((ABC)) = (6060) = 001090

⑥ Op code n i x b p e disp

000000 1 1 1 1 0 0 1010 1011 1100

03CABC

TA = disp + x + B = ABC + 0030 + 5000 = 5AEC

Value = (TA) = (5AEC) = 000030